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Copper-Catalyzed Coupling of Acyl Chlorides with *gem*-Difluorinated Organozinc Reagents via Acyl Dithiocarbamates

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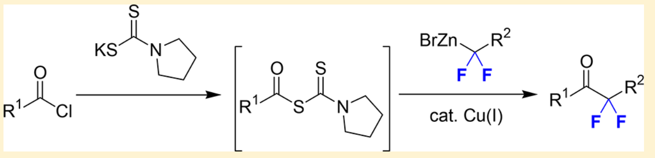
Supporting Information

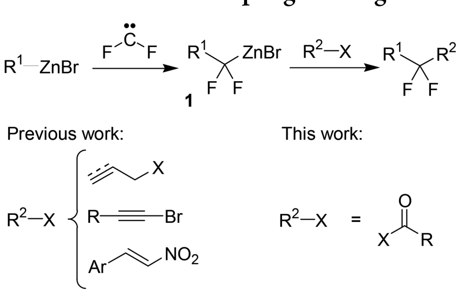
ABSTRACT: A cross-coupling of acyl chlorides with *gem*-difluorinated organozinc reagents affording difluorinated ketones is described. In the reaction, acyl chlorides are first treated with potassium dithiocarbamate to generate S-acyl dithiocarbamates, which couple with organozincs in the presence of a copper(I) catalyst.



Organofluorine compounds have found applications in various fields.[1] Molecules bearing fluorines at a specific position are of particular interest, primarily, in terms of their biological properties.[2] In this regard, methods allowing for the assembly of fluorinated compounds from simple building blocks have gained increasing attention in recent years.[3]

Recently, we described a family of *gem*-difluorinated organozinc reagents **1**, which are generated by the insertion of difluorocarbene into the carbon−zinc bond of conventional benzyl and alkyl organozincs[4] (Scheme 1). These reagents were

Scheme 1. Formation and Coupling of Reagents 1



treated with some heteroatom-based electrophiles[4,5] and involved in cross-couplings with allyl and propargyl halides,[6] 1-bromoalkynes,[7] and nitrostyrenes.[8] Herein we report on the interaction of organozincs **1** with acylating reagents.

It should be pointed out that the products of this coupling, α,α-difluoroketones, are useful for medicinal chemistry,[9] and several compounds of this type have been evaluated as drug candidates.[10] Indeed, owing to strong electron-withdrawing effect of fluorines, α,α-difluoroketones can easily add water at the carbonyl group, and the resulting adducts can mimic transition states of hydrolysis of the amide bond. Correspondingly, these ketones can serve as inhibitors of some proteolytic enzymes.[9,11]

Several approaches for the synthesis of difluorinated ketones have been reported,[12] but require either elevated temperatures or harsh chemical reagents. In this regard, a mild method based on the modification of derivatives of readily available carboxylic acids would be desirable. The acylation of difluorinated organolithiums has recently been evaluated, but required extremely low temperatures (−130 °C) owing to the instability of the latter reagents, and these reactions proceeded in low yields.[13] At the same time, fluorinated organozincs offer a nice combination of stability and reactivity.[14,15]

Couplings of nonfluorinated organozincs with derivatives of carboxylic acids are well-known. Typically, acyl chlorides (Negishi reaction)[16] and thioesters (Fukuyama reaction)[17] are employed as electrophilic partners in the presence of a transition-metal catalyst. For fluorinated organozincs, the couplings with acyl chlorides can be catalyzed by palladium(0) (for fluorinated vinyl organozincs[18]) or by copper(I) (for fluorinated phosphonates[19]), whereas uncatalyzed reactions are not productive.[20]

The reaction of various benzoyl derivatives with organozinc **1a** ($BnCF_2ZnBr$) originating from benzylzinc bromide[4] was evaluated (Table 1). Conventional acylating reagents such as benzoyl chloride and benzoic anhydride were completely ineffective when using palladium and copper catalysts (entries 1−4), and we switched to thioesters. The derivatives of ethanethiol, 2-pyridinethiol, 2-benzothiazolethiol, and 2-benzoxazolethiol provided moderate yields of the product (entries 5−12). As more reactive carbonyl electrophiles, acyl xanthates[21] and thiocarbamates[22] were considered. These species were generated in situ by the treatment of benzoyl chloride with readily available potassium xanthogenates or thiocarbamates. Finally, it was determined that pyrrolidine-substituted dithiocarbamate in combination with copper(I) chloride/triphenylphosphine complex ($CuCl\cdot1.5PPh_3$)[23] afforded product **2a** in 85% isolated yield (entry 20).[24] Among other catalysts, copper chloride bearing an imidazolidene ligand provided comparable

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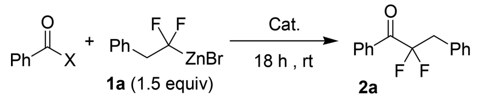
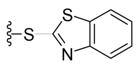
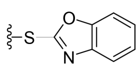
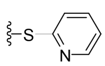
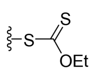
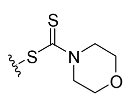
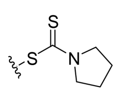
Table 1. Optimization Studies



No.	X	Cat. (mol %)[a]	Yield of **2a**, %[b]
1	Cl	$Pd(dba)_2$ (5)	-
2		$CuCl \cdot 1.5PPh_3$ (5)	-
3	OBz	$CuCl \cdot 1.5PPh_3$ (5)	-
4	CN	CuI (10)	-
5	SEt	CuI (10)	-
6		$Pd(dba)_2$ (5)	-
7		CuI (10)	53
8		CuI (20)	53
9		CuTC (10)	43
10		$CuCl \cdot 1.5PPh_3$ (10)	18
11		$CuCl \cdot 1.5PPh_3$ (5)	6
12		CuI (10)	-
13		CuI (10)	24
14		CuI (10)	45
15		CuI (10)	64
16		$Cu(OAc)_2$ (10)	12
17		CuCN (5)	68
18		Cu-Imes (5)	74
19		$CuCl \cdot 1.5PPh_3$ (5)	76
20		$CuCl \cdot 1.5PPh_3$ (10)	89 (85)[c]
21[d]		$CuCl \cdot 1.5PPh_3$ (10)	34
22		-	-

[a]CuTC = copper(I) thiophene-2-carboxylate. Cu-Imes = chloro[1,3-bis(2,4,6-trimethylphenyl)imidazol-2-ylidene]copper(I). [b]Determined by the ^{19}F NMR of reaction mixtures. [c]Isolated yield. [d]Reaction time = 6 h.

results (see entries 18 and 19), but owing to its high cost, the phosphine complexed salt was used for further work.

It should be pointed out that both acyl dithiocarbamates and *gem*-difluorinated organozincs have a limited stability, and to achieve good yields, they were mixed at −25 °C followed by slow warming to room temperature. Thus, the optimized conditions involve the treatment of acyl chloride with potassium dithiocarbamate (1.1 equiv) for about 2 min in DMF at −25 °C with the subsequent addition of a solution of organozinc **1** (1.5 equiv) in 1,2-dimethoxyethane. Though 10 mol % of the copper catalyst was used for aromatic acyl chlorides, for reactions of the derivatives of aliphatic acids, 20% of the catalyst gave higher yields. With this protocol, a series of difluorinated ketones **2** were obtained (Table 2). Aromatic, α,β-unsaturated, and α-unbranched aliphatic acyl chlorides provided good yields of products, whereas for derivatives of isobutyric and acetic acids, decreased yields were achieved (entries 11 and 12). The difluorinated ketone obtained from 4-cyanobenzoyl chloride was difficult to isolate, likely, because of propensity to the addition of water at the C═O bond; in this case, the crude product was reduced with sodium borohydride to the corresponding alcohol **2d**-OH (entry 3). The propensity of *gem*-difluorinated organozincs to decompose in the presence of copper(I) salts[6a] may be responsible for the decreased yields in some examples, as is the case for ester-substituted reagent **1g**.

In regard to the reaction mechanism, it can be proposed that a typical Cu(I)/Cu(III) catalytic cycle is realized[25] (Scheme 2). Thus, the reaction likely starts from the zinc/copper exchange leading to the moderately stable organocopper reagent **3**. Its subsequent oxidative addition into the carbon−sulfur bond of acyl dithiocarbamate generates the short-lived copper(III) intermediate **4**.[26] In the latter species, the copper may have an additional stabilizing interaction with the thioamide fragment typical for copper dithiocarbamates.[27] At the final step, reductive elimination from copper(III) affords the product and regenerates the copper(I) catalyst.

In an attempt to observe copper(I) complex **3**, several experiments were performed using the model reagent **1a**. Thus, when **1a** was combined with 1 equiv of $CuCl \cdot 1.5PPh_3$ in the reaction solvent system (DME/DMF), after 30 min at room temperature, 70% of reagent **1a** decomposed.[28] At the same time, no fluorinated organocopper species were observed by ^{19}F NMR. Interestingly, almost the same decomposition rate of **1a** was noted when only 0.1 equiv of $CuCl \cdot 1.5PPh_3$ was used.

In summary, a method for the synthesis of α,α-difluorinated ketones from acyl chlorides and difluorinated organozincs is described. Acyl dithiocarbamates are believed to serve as key electrophilic intermediates for the copper catalyzed cross-coupling. Since starting fluorinated organozincs are prepared from conventional organozincs and a difluorocarbene source, the method corresponds to the assembling of difluorinated ketones from three components, a nucleophilic reagent, difluorocarbene, and acyl electrophile.

■ EXPERIMENTAL SECTION

Dimethylformamide was distilled under a vacuum from P_2O_5 and stored over 4 Å molecular seives. High-resolution mass spectra (HRMS) were measured using electrospray ionization (ESI) and a time-of-flight (TOF) mass analyzer. The measurements were done in a positive ion mode (interface capillary voltage −4500 V) or in a negative ion mode (3200 V); the mass range was 50−3000 m/z . Column chromatography was carried out employing silica gel 230−400 mesh. Precoated silica gel plates F-254 were used for thin-layer analytical chromatography visualizing with UV and/or an acidic aqueous $KMnO_4$ solution. The copper complex $CuCl \cdot 1.5PPh_3$ was prepared according to a literature procedure.[29]

Synthesis of Ketones 2 (General Procedure). *Preparation of Reagents 1.* A solution of an organozinc reagent in THF (10 mmol)[4a]

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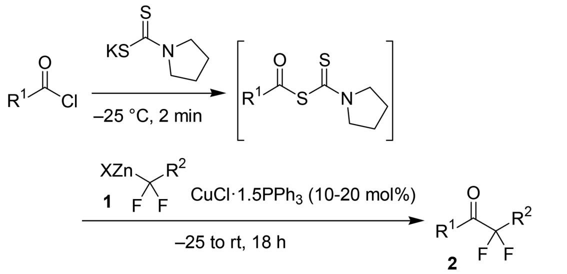
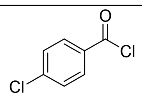
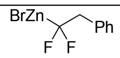
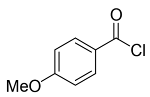
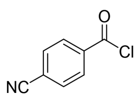
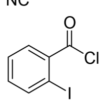
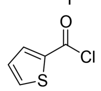
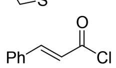
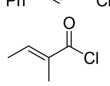
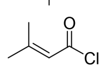
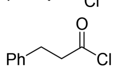
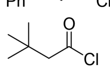
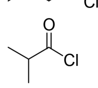
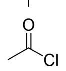
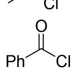
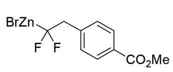
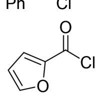
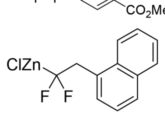
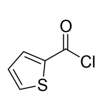
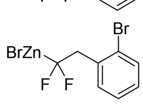
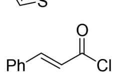
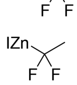
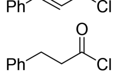
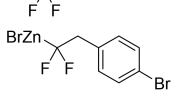
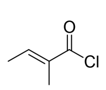
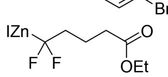
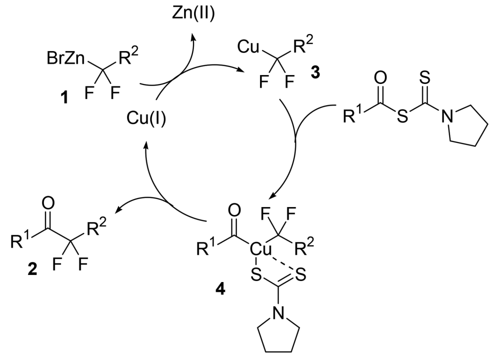
Table 2. Synthesis of Difluorinated Ketones 1



No.	Acyl chloride	Organozinc		2	Y., %[a]
1			**1a**	**2b**	83
2			**1a**	**2c**	80
3			**1a**	**2d**-OH[b]	73
4			**1a**	**2e**	71
5			**1a**	**2f**	75
6			**1a**	**2g**	68
7			**1a**	**2h**	75
8			**1a**	**2i**	74
9			**1a**	**2j**	70
10			**1a**	**2k**	70
11			**1a**	**2l**	55
12			**1a**	**2m**	48
13			**1b**	**2n**	75
14			**1c**	**2o**	75
15			**1d**	**2p**	63
16			**1e**	**2q**	68
17			**1f**	**2r**	62
18			**1g**	**2s**	46

[a]Isolated yield. [b]Alcohol obtained by reduction of the ketone.

Scheme 2. Proposed Mechanism



was added to dry potassium acetate (1176 mg, 12 mmol) at room temperature. The mixture was concentrated under a vacuum, and the residue was dissolved in 1,2-dimethoxyethane (15 mL). The mixture was cooled to −25 °C; Me_3SiCF_2Br (2440 mg, 12 mmol) was added dropwise, and the mixture was stirred at −25 °C for 18 h. The concentration of reagents **1** was determined by ^{19}F NMR with an internal standard.

*Coupling of Acyl Chlorides with Reagents **1**.* Acyl chloride (1 mmol) was added to a solution of potassium 1-pyrrolidinecarbodithioate[30] (204 mg, 1.1 mmol) in DMF (0.75 mL) at −25 °C, and the mixture was stirred for 2 min at −25 °C. Then, a solution of reagent **1** (1.5 mmol) and $CuCl·1.5PPh_3$ (for **2a–f,n–p**, 49.2 mg, 0.1 mmol; for **2g–m,q–s**, 98.4 mg, 0.2 mmol) was added. The reaction mixture was allowed to warm to room temperature in 6 h and was stirred for an additional 12 h. For the work up, water (7 mL) was added, and the mixture was extracted with methyl *tert*-butyl ether/hexane, (1/1, 3 × 5 mL). The combined organic layers were filtered through Na_2SO_4 and concentrated under a vacuum, and the residue was purified by column chromatography.

*2,2-Difluoro-1,3-diphenylpropan-1-one (**2a**).*[31] Yield 209 mg (85%). Colorless solid. Mp 45–46 °C. R_f 0.30 (hexane/ethyl acetate, 15/1). ^1H NMR (300 MHz, $CDCl_3$), δ: 8.08 (d, *J* = 7.6 Hz, 2H), 7.63 (t, *J* = 7.4 Hz, 1H), 7.49 (t, *J* = 7.6 Hz, 2H), 7.32–7.39 (m, 5H), 3.57 (t, *J* = 17.8 Hz, 2H). $^{13}C\{^1H\}$ NMR (75 MHz, $CDCl_3$), δ: 189.6 (t, *J* = 31.1 Hz), 134.3, 132.3 (t, *J* = 2.6 Hz), 131.3 (t, *J* = 3.7 Hz), 130.9, 130.2 (t, *J* = 3.4 Hz), 128.7, 128.5, 127.6, 118.5 (t, *J* = 254.7 Hz), 40.3 (t, *J* = 23.3 Hz). ^{19}F NMR (282 MHz, $CDCl_3$), δ: −99.3 (t, *J* = 17.8 Hz).

*1-(4-Chlorophenyl)-2,2-difluoro-3-phenylpropan-1-one (**2b**).* Yield 233 mg (83%). Colorless solid. Mp 46–47 °C. R_f 0.45 (hexane/ethyl acetate, 13/1). ^1H NMR (300 MHz, $CDCl_3$), δ: 7.98 (d, *J* = 8.8 Hz, 2H), 7.44 (d, *J* = 8.8 Hz, 2H), 7.34 (s, 5H), 3.54 (t, *J* = 17.8 Hz, 2H). $^{13}C\{^1H\}$ NMR (75 MHz, $CDCl_3$), δ: 188.5 (t, *J* = 31.6 Hz), 141.0, 131.6 (t, *J* = 3.5 Hz), 131.1 (t, *J* = 3.7 Hz), 130.9, 130.5 (t, *J* = 2.6 Hz), 129.0, 128.5, 127.7, 118.4 (t, *J* = 254.3 Hz), 40.1 (t, *J* = 23.1 Hz). ^{19}F NMR (282 MHz, $CDCl_3$), δ: −99.3 (t, *J* = 17.8 Hz). HRMS (ESI): calcd for $C_{15}H_{12}ClF_2O$ [M + H], 281.0539; found 281.0535.

*2,2-Difluoro-1-(4-methoxyphenyl)-3-phenylpropan-1-one (**2c**).* Yield 221 mg (80%). Colorless solid. Mp 70–71 °C. R_f 0.32 (hexane/ethyl acetate, 8/1). ^1H NMR(300 MHz, $CDCl_3$), δ: 8.05 (d, *J* = 9.0 Hz, 2H), 7.32 (s, 5H), 6.94 (d, *J* = 9.0 Hz, 2H), 3.89 (s, 3H), 3.51 (t, *J* = 18.0 Hz, 2H). $^{13}C\{^1H\}$ NMR (75 MHz, $CDCl_3$), δ: 188.0 (t, *J* = 30.7 Hz), 164.5, 132.8 (t, *J* = 3.4 Hz), 131.6 (t, *J* = 3.7 Hz), 130.9, 128.5, 127.6, 125.1 (t, *J* = 2.9 Hz), 118.7 (t, *J* = 254.7 Hz), 114.0, 55.6, 40.4 (t, *J* = 23.4 Hz). ^{19}F NMR(282 MHz, $CDCl_3$), δ: −99.1 (t, *J* = 18.0 Hz). HRMS (ESI): calcd for $C_{16}H_{14}F_2O_2Na$ [M + Na], 299.0854; found, 299.0858.

*4-(2,2-Difluoro-1-hydroxy-3-phenylpropyl)benzonitrile (**2d**–OH).* The crude product was passed through a silica gel column eluting with hexane/EtOAc, 5/1. After evaporation of the solvent, the residue was dissolved in methanol (3 mL). The mixture was cooled to 0 °C, and sodium borohydride (43 mg, 1.13 mmol) was added. The cooling bath was removed, and the mixture was stirred for 4 h at room temperature.

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For the work up, water (5 mL) was added, and the mixture was extracted with methyl *tert*-butyl ether (3 × 5 mL). The combined organic layers were filtered through Na_2SO_4 and concentrated under a vacuum, and the residue was purified by column chromatography on silica gel eluting with hexane/ethyl acetate, 5/2 (R_f 0.45). Yield 200 mg (73%). Colorless solid. Mp 117−118 °C. ^{1}H NMR (300 MHz, $CDCl_3$), δ: 7.67 (d, *J* = 8.1 Hz, 2H), 7.59 (d, *J* = 8.1 Hz, 2H), 7.40−7.25 (m, 5H), 4.86 (ddd, *J* = 13.3, 6.1, 5.9 Hz, 1H), 3.39 (ddd, *J* = 22.3, 16.1, 14.7 Hz, 1H), 3.07 (ddd, *J* = 22.3, 14.7, 9.4 Hz, 1H), 3.05−2.96 (m, 1H). ^{13}C{^{1}H} NMR (75 MHz, $CDCl_3$), δ: 142.1, 132.2 (d, *J* = 4.9 Hz), 132.0, 130.6, 128.5, 127.6, 122.0 (dd, *J* = 249.4, 246.5 Hz), 118.7, 112.2, 73.6 (dd, *J* = 30.5, 27.9 Hz), 38.8 (t, *J* = 24.2 Hz). ^{19}F NMR (282 MHz, $CDCl_3$), δ: 106.3 (dddd, *J* = 252.2, 22.3, 16.1, 6.1 Hz, 1F), 109.6 (dddd, *J* = 252.2, 22.3, 13.3, 9.4 Hz, 1F). HRMS (ESI): calcd for $C_{16}H_{13}F_2NONa$ [M + Na], 296.0857; found 296.0862.

2,2-Difluoro-1-(2-iodophenyl)-3-phenylpropan-1-one (**2e**). Yield 264 mg (71%). Yellow oil. R_f 0.30 (hexane/ethyl acetate, 25/1). ^{1}H NMR (300 MHz, $CDCl_3$), δ: 7.95 (d, *J* = 8.0 Hz, 1H), 7.42−7.26 (m, 7H), 7.16 (ddd, *J* = 8.0, 7.2, 2.1 Hz, 1H), 3.58 (t, *J* = 17.2 Hz, 2H). ^{13}C{^{1}H} NMR (75 MHz, $CDCl_3$), δ: 193.8 (t, *J* = 32.3 Hz), 140.9, 138.9 (t, *J* = 1.6 Hz), 132.6, 131.1, 130.8 (t, *J* = 3.9 Hz), 128.8 (t, *J* = 4.1 Hz), 128.6, 127.9, 127.6, 117.4 (t, *J* = 255.4 Hz), 92.6, 39.8 (t, *J* = 23.3 Hz). ^{19}F NMR (282 MHz, $CDCl_3$), δ: −99.9 (t, *J* = 17.2 Hz). HRMS (ESI): calcd for $C_{15}H_{11}F_2IONa$ [M + Na], 394.9715; found, 394.9701.

2,2-Difluoro-3-phenyl-1-(thiophen-2-yl)propan-1-one (**2f**). Yield 189 mg (75%). Colorless oil. R_f 0.30 (hexane/ethyl acetate, 15/1). ^{1}H NMR (300 MHz, $CDCl_3$), δ: 7.95−7.92 (m, 1H), 7.76 (d, *J* = 4.8 Hz, 1H), 7.40−7.32 (m, 5H), 7.17 (t, *J* = 4.8 Hz, 1H), 3.55 (t, *J* = 17.4 Hz, 2H). ^{13}C{^{1}H} NMR (75 MHz, $CDCl_3$), δ: 182.9 (t, *J* = 31.7 Hz), 138.4 (t, *J* = 2.7 Hz), 136.5, 135.9 (t, *J* = 5.4 Hz), 131.1 (t, *J* = 3.6 Hz), 130.9, 128.8, 128.5, 127.8, 118.2 (t, *J* = 254.3 Hz), 40.4 (t, *J* = 23.5 Hz). ^{19}F NMR (282 MHz, $CDCl_3$), δ: −100.7 (t, *J* = 17.4 Hz). HRMS (ESI): calcd for $C_{13}H_{10}F_2OSNa$ [M + Na], 275.0313; found, 275.0312.

(E)-4,4-Difluoro-1,5-diphenylpent-1-en-3-one (**2g**). Yield 185 mg (68%). Colorless solid. Mp 58−59 °C. R_f 0.33 (hexane/ethyl acetate, 15/1). ^{1}H NMR (300 MHz, $CDCl_3$), δ: 7.85 (d, *J* = 15.9 Hz, 1H), 7.60 (dd, *J* = 7.6, 1.7 Hz, 2H), 7.50−7.28 (m, 8H), 7.04 (d, *J* = 15.9 Hz, 1H), 3.47 (t, *J* = 17.2 Hz, 2H). ^{13}C{^{1}H} NMR (75 MHz, $CDCl_3$), δ: 189.3 (t, *J* = 30.5 Hz), 147.6, 134.0, 131.6, 131.2 (t, *J* = 3.9 Hz), 130.8, 129.1, 129.0, 128.5, 127.7, 118.1, 117.7 (t, *J* = 253.7 Hz), 39.8 (t, *J* = 23.7 Hz). ^{19}F NMR (282 MHz, $CDCl_3$), δ: −106.5 (t, *J* = 17.2 Hz). HRMS (ESI): calcd for $C_{17}H_{14}F_2ONa$ [M + Na], 295.0905; found, 295.0902.

(E)-2,2-Difluoro-4-methyl-1-phenylhex-4-en-3-one (**2h**). Yield 168 mg (75%). Colorless oil. R_f 0.29 (hexane/ethyl acetate, 22/1). ^{1}H NMR (300 MHz, $CDCl_3$), δ: 7.48−7.26 (m, 5H), 7.11 (q, *J* = 7.0 Hz, 1H), 3.46 (t, *J* = 17.7 Hz, 2H), 1.94 (d, *J* = 7.0 Hz, 3H), 1.88 (s, 3H). ^{13}C{^{1}H} NMR (75 MHz, $CDCl_3$), δ: 190.3 (t, *J* = 29.2 Hz), 143.7 (t, *J* = 6.1 Hz), 133.7, 131.9 (t, *J* = 3.7 Hz), 130.9, 128.4, 127.5, 118.6 (t, *J* = 255.6 Hz), 40.8 (t, *J* = 23.6 Hz), 15.3, 11.5. ^{19}F NMR (282 MHz, $CDCl_3$), δ: −97.7 (t, *J* = 17.7 Hz). HRMS (ESI): calcd for $C_{13}H_{14}F_2ONa$ [M + Na], 247.0905; found, 247.0904.

2,2-Difluoro-5-methyl-1-phenylhex-4-en-3-one (**2i**). Yield 166 mg (74%). Colorless oil. R_f 0.30 (hexane/ethyl acetate, 18/1). ^{1}H NMR (300 MHz, $CDCl_3$), δ: 7.44−7.25 (m, 5H), 6.43−6.34 (m, 1H), 3.38 (t, *J* = 17.2 Hz, 2H), 2.24 (d, *J* = 0.9 Hz, 3H), 1.98 (d, *J* = 0.9 Hz, 3H). ^{13}C{^{1}H} NMR (75 MHz, $CDCl_3$), δ: 189.4 (t, *J* = 28.4 Hz), 164.2, 131.6 (t, *J* = 3.8 Hz), 130.7, 128.5, 127.6, 117.5 (t, *J* = 254.8 Hz), 117.2, 39.6 (t, *J* = 23.9 Hz), 28.5, 21.7. ^{19}F NMR (282 MHz, $CDCl_3$), δ: −106.9 (t, *J* = 17.2 Hz). HRMS (ESI): calcd for $C_{13}H_{18}F_2ON$ [M + NH_4], 242.1351; found, 242.1352.

2,2-Difluoro-1,5-diphenylpentan-3-one (**2j**). Yield 192 mg (70%). Colorless oil. R_f 0.42 (hexane/ethyl acetate, 15/1). ^{1}H NMR (300 M/Hz, $CDCl_3$), δ: 7.45−7.10 (m, 10H), 3.38 (t, *J* = 16.9 Hz, 2H), 2.95−2.80 (m, 4H). ^{13}C{^{1}H} NMR (75 MHz, $CDCl_3$), δ: 200.8 (t, *J* = 31.1 Hz), 140.2, 130.9 (t, *J* = 4.6 Hz), 130.6, 128.6, 128.5, 128.3, 127.8, 126.4, 117.3 (t, *J* = 254.5 Hz), 39.5 (t, *J* = 23.5 Hz), 39.0, 28.5. ^{19}F NMR (282 MHz, $CDCl_3$), δ: −106.3 (t, *J* = 16.9 Hz). HRMS (ESI): calcd for $C_{17}H_{16}F_2ONa$ [M + Na], 297.1061; found, 297.1070.

2,2-Difluoro-5,5-dimethyl-1-phenylhexan-3-one (**2k**). Yield 168 mg (70%). Colorless oil. R_f 0.50 (hexane/ethyl acetate, 20/1). ^{1}H NMR (300 MHz, $CDCl_3$), δ: 7.43−7.21 (m, 5H), 3.33 (t, *J* = 16.9 Hz, 2H), 2.41 (s, 2H), 1.01 (s, 9H). ^{13}C{^{1}H} NMR (75 MHz, $CDCl_3$), δ: 201.0 (t, *J* = 30.1 Hz), 131.3 (t, *J* = 4.2 Hz), 130.9, 128.6, 127.8, 117.0 (t, *J* = 255.1 Hz), 48.7, 39.3 (t, *J* = 23.7 Hz), 30.8, 29.4. ^{19}F NMR (282 MHz, $CDCl_3$), δ: −106.2 (t, *J* = 16.9 Hz). HRMS (ESI): calcd for $C_{14}H_{22}F_2ON$ [M + NH_4], 258.1664; found, 258.1665.

2,2-Difluoro-4-methyl-1-phenylpentan-3-one (**2l**). Yield 117 mg (55%). Colorless oil. Chromatography: hexane/ethyl acetate, 15/1. ^{1}H NMR (300 MHz, $CDCl_3$), δ: 7.42−7.22 (m,5H), 3.36 (t, *J* = 16.9 Hz, 2H), 2.93 (sept, *J* = 6.8 Hz, 1H), 1.03 (d, *J* = 6.8 Hz, 6H). ^{13}C{^{1}H} NMR (75 MHz, $CDCl_3$), δ: 205.4 (t, *J* = 29.7 Hz), 131.2 (t, *J* = 4.4 Hz), 130.8, 128.6, 127.7, 117.9 (t, *J* = 255.0 Hz), 39.7 (t, *J* = 23.6 Hz), 35.6, 17.8. ^{19}F NMR (282 MHz, $CDCl_3$), δ: −105.4 (t, *J* = 16.9 Hz). HRMS (ESI): calcd for $C_{12}H_{14}F_2ONa$ [M + Na], 235.0905; found, 235.0912.

3,3-Difluoro-4-phenylbutan-2-one (**2m**). Yield 88 mg (48%). Colorless oil. Chromatography: hexane/ethyl acetate, 20/1. ^{1}H NMR (300 MHz, $CDCl_3$), δ: 7.51−7.14 (m, 5H), 3.34 (t, *J* = 17.0 Hz, 2H), 2.19 (t, *J* = 1.4 Hz, 3H). ^{13}C{^{1}H} NMR (75 MHz, $CDCl_3$), δ: 199.3 (t, *J* = 32.2 Hz), 131.1 (t, *J* = 4.2 Hz), 130.7, 128.7, 127.8, 117.0 (t, *J* = 253.6 Hz), 39.1 (t, *J* = 23.5 Hz), 24.8. ^{19}F NMR (282 MHz, $CDCl_3$), δ: −105.9 (t, *J* = 17.0 Hz). HRMS (ESI): calcd for $C_{10}H_{10}F_2ONa$ [M + Na], 207.0592; found, 207.0590.

Methyl 4-(2,2-Difluoro-3-oxo-3-phenylpropyl)benzoate (**2n**). Yield 228 mg (75%). Colorless solid. Mp 114−115 °C. The crude product was purified by flash chromatography (hexane/ethyl acetate, 8/1), followed by purification by preparative HPLC (reversed-phase column C_{18}, 21.2 mm × 250 mm, 5 μm, flow rate 12 mL/min, 17% water in acetonitrile, retention time 9 min). ^{1}H NMR (300 MHz, $CDCl_3$), δ: 8.07 (d, *J* = 7.7 Hz, 2H), 8.01 (d, *J* = 7.8 Hz, 2H), 7.64 (t, *J* = 7.2 Hz, 1H), 7.54−7.31 (m, 4H), 3.92 (s, 3H), 3.58 (t, *J* = 17.6 Hz, 2H). ^{13}C{^{1}H} NMR (75 MHz, $CDCl_3$), δ: 189.1 (t, *J* = 32.4 Hz), 166.9, 136.7, 134.5, 132.0, 131.0, 130.2, 129.7, 129.6, 128.8, 118.3 (t, *J* = 257.0 Hz), 52.2, 40.1 (t, *J* = 23.3 Hz). ^{19}F NMR (282 MHz, $CDCl_3$), δ: −98.9 (t, *J* = 17.6 Hz). HRMS (ESI): calcd for $C_{17}H_{15}F_2O_3$ [M + H], 305.0984; found, 305.0984.

2,2-Difluoro-1-(furan-2-yl)-3-(naphthalen-1-yl)propan-1-one (**2o**). Yield 214 mg (75%). Colorless oil. R_f 0.32 (hexane/ethyl acetate, 5/1). ^{1}H NMR (300 MHz, $CDCl_3$), δ: 8.08 (d, *J* = 8.4 Hz, 1H), 7.92−7.77 (m, 2H), 7.67 (d, *J* = 1.5 Hz, 1H), 7.61−7.38 (m, 4H), 7.28−7.22 (m, 1H), 6.47 (dd, *J* = 3.7, 1.6 Hz, 1H), 3.99 (t, *J* = 17.1 Hz, 2H). ^{13}C{^{1}H} NMR (75 MHz, $CDCl_3$), δ: 178.2 (t, *J* = 31.6 Hz), 149.0, 148.4 (t, *J* = 1.9 Hz), 133.9, 132.9, 129.8, 128.7, 127.3 (t, *J* = 3.1 Hz), 126.4, 125.8, 125.2, 124.1 (t, *J* = 1.5 Hz), 123.3 (t, *J* = 5.9 Hz), 118.2 (t, *J* = 254.4 Hz), 112.8, 36.5 (t, *J* = 23.8 Hz). ^{19}F NMR (282 MHz, $CDCl_3$), δ −100.7 (t, *J* = 17.1 Hz). HRMS (ESI): calcd for $C_{17}H_{16}F_2NO_2$ [M + NH_4], 304.1144; found, 304.1149.

3-(2-Bromophenyl)-2,2-difluoro-1-(thiophen-2-yl)propan-1-one (**2p**). Yield 214 mg (65%). Colorless solid. Mp 57−58 °C. R_f 0.35 (hexane/ethyl acetate, 12/1). ^{1}H NMR (300 MHz, $CDCl_3$), δ: 7.99 (dd, *J* = 3.8, 1.2 Hz, 1H), 7.80 (dd, *J* = 4.9, 0.8 Hz, 1H), 7.60 (dd, *J* = 8.0, 0.8 Hz, 1H), 7.44 (d, *J* = 7.5 Hz, 1H), 7.31 (td, *J* = 7.5, 1.3 Hz, 1H), 7.23−7.12 (m, 2H), 3.76 (t, *J* = 17.7 Hz, 2H). ^{13}C{^{1}H} NMR (75 MHz, $CDCl_3$), δ: 182.6 (t, *J* = 31.3 Hz), 138.3 (t, *J* = 2.4 Hz), 136.7, 136.1 (t, *J* = 5.1 Hz), 133.2, 132.8, 131.4 (t, *J* = 2.6 Hz), 129.4, 128.9, 127.5, 126.1, 117.9 (t, *J* = 255.1 Hz), 39.6 (t, *J* = 23.3 Hz). ^{19}F NMR (282 MHz, $CDCl_3$), δ: −100.9 (t, *J* = 17.6 Hz). HRMS (ESI): calcd for $C_{13}H_{10}BrF_2OS$ [M + H], 330.9598, 332.9578; found 330.9594, 332.9572.

(E)-4,4-Difluoro-1-phenylpent-1-en-3-one (**2q**).[12c] Yield 133 mg (68%). Colorless oil. R_f 0.50 (hexane/ethyl acetate, 8/1). ^{1}H NMR (300 MHz, $CDCl_3$), δ: 7.90 (d, *J* = 16.0 Hz, 1H), 7.67−7.60 (m, 2H), 7.51−7.37 (m, 3H), 7.16 (dt, *J* = 15.9, 1.4 Hz, 1H), 1.80 (t, *J* = 19.2 Hz, 3H). ^{13}C{^{1}H} NMR (75 MHz, $CDCl_3$), δ: 189.0 (t, *J* = 31.5 Hz), 147.8, 134.1, 131.6, 129.2, 129.0, 118.3 (t, *J* = 249.1 Hz), 117.8, 19.9 (t, *J* = 25.2 Hz). ^{19}F NMR (282 MHz, $CDCl_3$), δ: −100.7 (q, *J* = 19.2 Hz).

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*1-(4-Bromophenyl)-2,2-difluoro-5-phenylpentan-3-one (**2r**).* Yield 218 mg (62%). Colorless oil. The crude product was purified by flash chromatography (hexane/ethyl acetate, 15/1), followed by purification by preparative HPLC (reversed-phase column C_{18}, 21.2 mm × 250 mm, 5 μm, flow rate 12 mL/min, 27% water in acetonitrile, retention time 15.5 min). ^{1}H NMR (300 MHz, $CDCl_3$), δ: 7.49−7.42 (m, 2H), 7.37−7.21 (m, 3H), 7.19−7.06 (m, 4H), 3.28 (t, *J* = 16.8 Hz, 2H), 2.92−2.85 (m, 4H). $^{13}C\{^1H\}$ NMR (75 MHz, $CDCl_3$), δ: 200.3 (t, *J* = 31.2 Hz), 140.1, 132.3, 131.8, 130.0 (t, *J* = 3.9 Hz), 128.7, 128.4, 126.5, 122.1, 116.8 (t, *J* = 254.3 Hz), 38.7, 38.6 (t, *J* = 23.5 Hz), 28.5. ^{19}F NMR (282 MHz, $CDCl_3$), δ: −106.2 (t, *J* = 16.8 Hz). HRMS (ESI): calcd for $C_{17}H_{15}BrF_2ONa$ [M + Na], 375.0167, 377.0147; found, 375.0163, 377.0144.

*(E)-Ethyl 5,5-Difluoro-7-methyl-6-oxonon-7-enoate (**2s**).* Yield 114 mg (46%). Colorless oil. R_f 0.30 (hexane/ethyl acetate, 10/1). ^{1}H NMR (300 MHz, $CDCl_3$), δ: 7.08 (qq, *J* = 7.0, 1.2 Hz, 1H), 4.13 (q, *J* = 7.1 Hz, 2H), 2.38 (t, *J* = 7.4 Hz, 2H), 2.24−2.00 (m, 2H), 1.92 (d, *J* = 7.0 Hz, 3H), 1.89−1.76 (m, 5H), 1.25 (t, *J* = 7.1 Hz, 3H). $^{13}C\{^1H\}$ NMR (75 MHz, $CDCl_3$), δ: 190.2 (t, *J* = 29.5 Hz), 172.8, 143.6 (t, *J* = 4.8 Hz), 133.4, 119.6 (t, *J* = 254.0 Hz), 60.4, 33.7 (t, *J* = 23.4 Hz), 33.6, 17.2 (t, *J* = 4.6 Hz), 15.3, 14.2, 11.4. ^{19}F NMR (282 MHz, $CDCl_3$), δ: −99.1 (t, *J* = 17.5 Hz). HRMS (ESI): calcd for $C_{12}H_{19}F_2O_3$ [M + H], 249.1297; found, 249.1299.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.joc.7b02598.

Copies of NMR spectra for all compounds (PDF)

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Notes

The authors declare no competing financial interest.

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